Saima

Edmonton Alberta, Canada

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EDUCATION

Postdoctoral Scholar

University of Alberta, Edmonton, Canada

Project: Effect of Suforaphane (SFA) and ketogenic diet in pediatrics gliomas therapeutics

Advisor: Dr. Sujata Persad

Doctor of Philosophy in Molecular Genetics

COMSATS University Islamabad, Pakistan

CGPA: 3.65/4.0 (91%; letter grade: A-)

Thesis: Genetic Analysis of Ciliopathies Disease Spectrum

Advisor: Dr. Maleeha Azam

Masters in Pediatrics

University of Alberta, Edmonton, Canada

CGPA: 3.7/4.0 (92.5%; letter grade: A-)

Thesis: Overexpression of Active Beta Catenin in Osteosarcoma and its Metastatic Effect

Advisor: Dr. Sujata Persad

Master of Science in Biotechnology and Bioinformatics

Balochistan University Information Technology Engineering and Management Sciences, Pakistan

CGPA: 3.47/4.0 (86%, letter grade: A-)

Thesis: Molecular Genetic Analysis of Inherited Nail Dysplasia

Advisor: Dr. Rozeena Sheikh

Bachelor of Science in Biotechnology and Biochemistry

University of Balochistan, Pakistan

CGPA: 3.74/4.0 (94%, letter grade: A)

WORK EXPERIENCE

Postoctoral Scholar, University of Alberta, Canada (09/2025-present)

- Primary project: Understanding the mechanism of pediatrics glioblastoma
- Mentoring: Mentoring a CELL499 student.
- Lab management: Managing the lab independently.

Graduate Research Assistant Fellow, University of Alberta, Canada

- Primary project: Understanding the mechanism of osteosarcoma
- Mentoring: Mentored summer students and a CELL499 student.

Research Student, COMSATS University Islamabad, Pakistan

- Primary project: Molecular analysis of ciliopathies disease spectrum
- Mentoring: Mentored 5 masters and 4 undergraduate students
- Lab management: Organizing the lab instruments and inventory.
- Other: Establishing new protocols and writing down standard operating procedures

Visiting Research Student, University of Alberta, Canada

- Primary project(s): *CYP26B1* expression plasmid, exome sequencing for rare skeletal deformities
- Techniques utilized: genome/exome/sanger sequence analyses, functional evaluation, molecular cloning, expression analysis, western blotting.

RESEARCH ARTICLES

1. Hinton K, Ghafoor S, Landry T, Garcia E, Bush J, Asgarali R, Hitt M, Paul P, Eisenstat D. D, Persad S. Active Beta-Catenin (ABC) promotes an invasive phenotype in osteosarcoma. Scientific reports. DOI:10.21203/rs.3.rs-4979798/v1.*(First co-author).*
2. Ghafoor S, Garcia E, Daniel J. Jay, Persad S. Molecular Mechanism in promoting Epithelial Mesenchymal Transition (EMT). doi: 10.3390/ijms26094364.
3. Silveira K.d.C, Fonseca I. C, Oborn C, Wengryn P, Ghafoor S, Beke A, Dreseris E. S, Wong C, Iacovone A, Soltys C. L, 1, Babul-Hirji R, Artigalas O, Tavares A. A, Gingras A. C, Campos E, Cavalcanti D. P, Kannu P. CYP26B1-related disorder: expanding the ends of the spectrum through clinical and molecular evidence. Hum Genet 2023 Sep 27. https://doi: 10.1007/s00439-023-02598-2.
4. Ghafoor S, Silveira K.d.C, Qamar, R, Azam M, Kannu P. Exome Sequencing Identifies a Biallelic

GALNS Variant (p.Asp233Asn) Causing Mucopolysaccharidosis Type IVA in a Pakistani Consanguineous Family. Genes 2022, 13, 1743. https://doi.org/ 10.3390/genes13101743.

5. Ghafoor S, Rafique A, Shah S.T.B, Ansar M, Paton T, Ajmal M, Agha Z, Qamar R, Azam M. *KIF1A* novel frameshift variant p.(Ser887Profs*64) exhibits clinical heterogeneity in a Pakistani family with hereditary sensory and autonomic neuropathy type IIC. Int. J. of Neuroscience 2022, 1-17. https://doi.org/10.1080/00207454.2022.2140428.
6. Ghafoor S, Silveira K, Qamar R, Azam M, and Kannu P. A case report of juvenile ALS in a Pakistani Consanguineous family. *Manuscript under preparation.*
7. Ghafoor S, Qamar R, Azam M. Whole genome sequencing identifies pathogenic variants in *APTX* and *MPZ* genes causing Ataxia and Charcot Marie Tooth disease in two Pakistani Consanguineous families. *Manuscript under preparation.*

ABSTRACTS

1. Ghafoor S, Garcia E, Daniel J. Jay, Persad S. Role of Active Beta Catenin in Osteosarcoma Invasion & Metastasis. Cancer Research Institute of Northern Alberta (CRINA) 2025, November 12, 2025, Canada.
2. Ghafoor S, Garcia E, Daniel J. Jay, Persad S. Role of Active Beta Catenin in Osteosarcoma Invasion & Metastasis Canadian Cancer Research Conference (CCRC) 2025, November 2 to 4, 2025, Canada.
3. Ghafoor S, Garcia E, Daniel J. Jay, Persad S. Role of Active Beta Catenin in Osteosarcoma Invasion & Metastasis. CRINA research day 2024, November 13, 2024. University of Alberta, Canada.
4. Ghafoor S, Hinton K, Garcia E, Persad S. Overexpression of Active Beta-Catenin Promotes an Invasive Phenotype in Osteosarcoma. Pediatrics Research day 2024, April 17, 2024. University of Alberta, Canada.
5. Silveira K.d.C, Fonseca I. C, Oborn C, Ghafoor S, Artigalas O, Tavares A. A, Gingras A. C, Campos E, Cavalcanti D. P, Kannu P. CYP26B1 involvement in retinoic acid catabolism associated skeletal dysplasias. ISDS 2022 skeletal dysplasia meeting, August 24-27, Santiago, Chile.
6. Ghafoor S, Silveira K, Azam M, Kannu P. Genetic analysis of skeletal disorders in a Pakistani population through whole exome sequencing. WCHRI Research Day 2021, November 3–4, 2021. University of Alberta, Canada.

7. Ghafoor S, Silveira K, Azam M, Kannu P. Genetic analysis of skeletal disorders in a Pakistani population through whole exome sequencing. Medical Genetics research day 2021, October 20, 2021. University of Alberta, Canada.

CERTIFICATION AND RECOGNITIONS

- Certificate of Analytical X-Ray Equipment Radiation Safety Training (University ofAlberta, November, 2025)
- Graduate Embedded Certificate in Maternal and Child Health Research (University of Alberta, November, 2025)
- Certificate of Mouse Isoflurane Anesthesia (University ofAlberta, January 2024)
- Certificate of Subcutaneous Injection (University ofAlberta, December 2023)
- Certificate of Introduction to Basic Surgical Techniques in Research (University ofAlberta, December 2023)
- Certificate of Aseptic Techniques in Surgery (University ofAlberta, November 2023)
- Certificate of Laser Safety (University ofAlberta, September 2023)
- Certificate of Care and Use of the Mouse in Research (University ofAlberta, September 2023)
- Certificate of Basic Principles of Animal Ethics and Welfare of Animal Use (University of Alberta, 2021)
- Certificate of Supervisory HSF Professional Development (University of Alberta, 2021)
- Certificate of Workplace Hazardous Materials Information System (University of Alberta, 2021)
- Certificate of Laboratory Safety (University of Alberta, 2021)
- Certificate of Concept in Biosafety (University of Alberta, 2021)
- Certificate of participation in two days international training workshop on scientific paper and patent writing (Islamic World Educational, Scientific and Cultural Organization ICESCO, 2020)
- Advanced Courses from DELTA Academy for English reading, writing, and speaking skills (Dynamic English Language and Teaching Academy, Turbat, Pakistan, 2016)
- One-week training workshop on “Research Writing Skills” by BUITEMS Quetta (BUITEMS, 2015)

LANGUAGES

English (proficient), Urdu (proficient), Balochi (native)

AWARDS AND ACHIEVEMENTS

- Cancer Research Institute of Northern Alberta (CRINA) travel award, (University of Alberta, 2025)
- Graduate Students' Association (GSA) travel award, (University of Alberta, 2025)
- Medical Sciences Graduate Program Scholarship (University of Alberta, 2024)
- Alberta Graduate Excellence Scholarship (Government of Alberta, 2023)
- The University of Alberta Graduate Recruitment scholarship (University of Alberta, 2023)
- Maternal and Child Health program (MatCH) scholarship (University of Alberta, 2023)
- International Research Fellowship (Higher EducationCommission Pakistan, 2021)
- Prime Minister's Laptop Award" (Higher Education Commission Pakistan, 2014)
- Indigenous Scholarship for MS leading to Ph.D. (Higher Education Commission Pakistan, 2013)

REFERENCES

References available upon request.